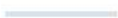


Trees, Ensembles and Imbalanced Data

Build calibrated classification workflows and decision thresholds.



Learning outcomes

01

Compare ensembles with a linear baseline.

02

Evaluate rare events with suitable metrics.

03

Keep weighting or resampling inside training folds.

The concept system

01

decision trees

02

ensembles

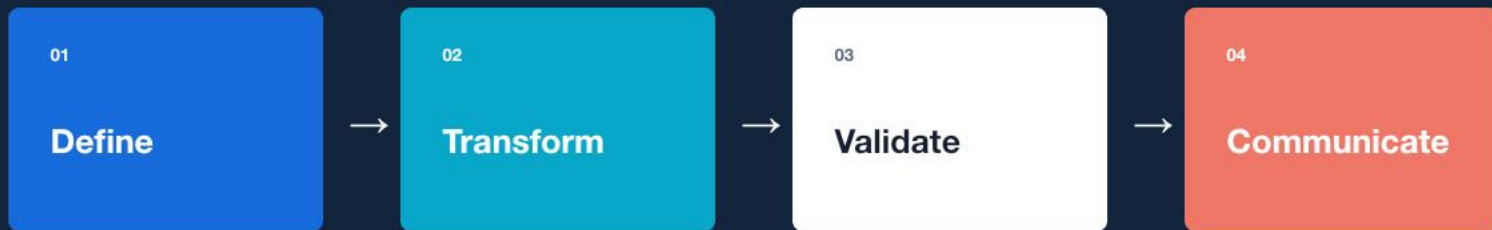
03

class imbalance

04

precision and recall

A repeatable reasoning loop



Detect rare payment failures and inspect false alarms.

Guided lab

LAB BRIEF

**Detect rare payment failures
and inspect false alarms.**

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

An imbalanced-classification report and operating point.

QUALITY GATE

- Reproducible
- Tested
- Decision-relevant

WEEKLY ASSESSMENT

Rare-Event Classifier

Leakage 25%, metrics 25%, threshold 20%, calibration 15%, communication 15%.

What should remain after the lesson?

CONCEPT

decision trees + ensembles

PRACTICE

Detect rare payment failures and inspect false alarms.

EVIDENCE

An imbalanced-classification report and operating point.
